(26) $\frac{1}{6!} + \frac{1}{7!} = \frac{x}{8!}$ find x

(27) $\frac{2x-1}{3} \geq \left(\frac{3x-2}{4}\right) - \left(\frac{2-x}{5}\right)$ solve for real x

(28) solve system of linear inequalities

$3x - 7 < 5 + x$

$11 - 5x \leq 1$

To prove

(29) $\cos\left(\frac{3\pi}{4} + x\right) - \cos\left(\frac{3\pi}{4} - x\right) = -\sqrt{2}\sin x$

(30) To prove

$$\frac{\cos 4x + \cos 3x + \cos 2x}{\sin 4x + \sin 3x + \sin 2x} = \cot 3x$$

(31) $A = \{1, 2, 3, 4, 6\}$ let R be relation on A defined by

$\{(a, b) : a, b \in A$, b is exactly divisible by a

(i) write R in roster form

(ii) find domain of R

(iii) find Range of R

(32) If $A \cup B = A \cup C$ and $A \cap B = A \cap C$

Show $B = C$

(33) $U = \{1, 2, 3, 4, 5, 6\}$ $A = \{2, 3\}$

$B = \{3, 4, 5\}$

Prove demorgan law by above example

(34) $A = \{1, 2, \{3, 4\}, 5\}$ which are incorrect & why

(i) $\{3, 4\} \in A$ (ii) $\{\{3, 4\}\} \subset A$

(iii) $1 \in A$ (iv) $1 \subset A$ (v) $\{1, 2, 5\} \subset A$

(vi) $\phi \subset A$ (viii) $\{1, 2, 3\} \subset A$ (ix) $\phi \in A$

(35) Draw Venn diagram of following

(i) $(A \cup B)'$ (ii) $(A \cap B)'$

(36) To prove

$$(\cos x + \cos y)^2 + (\sin x - \sin y)^2 = 4\cos^2 \frac{x+y}{2}$$

(37) Prove $2\sin^2 \frac{3\pi}{4} + 2\cos^2 \frac{\pi}{4} + 2\sec^2 \frac{\pi}{3} = 10$

(38) If in two circle arcs of same length subted angles 60° and 75° at centre find ratio of their radii

(39) $2\,{}^nC_3 : {}^nC_3 = 12 : 1$ find n

(40) what is number of ways of chosing 4 cards from a pack of 52 playing cards.

(i) four cards are of same suit
(ii) four cards belong to 4 different suits
(iii) are face cards
(iv) two are red card and 2 are black cards
(v) cards are of same colour

(41) to prove $\cos 6x = 32\cos^6 x - 48\cos^4 x + 18\cos^2 x - 1$

(42) show
$\tan 3x \cdot \tan 2x \cdot \tan x = \tan 3x - \tan 2x - \tan x$

(43) In how many ways can 5 girls and 3 boys be seated in a row so that no 2 boys are together

(44) Determine the number of 5 cards combinations out of 52 card if selection of 5 cards has exactly one king.

(45) find a) $\operatorname{cosec}(-1410^\circ)$ b) $\tan \frac{19\pi}{3}$

c) $\cot\left(-\frac{15\pi}{4}\right)$ d) $\cos(-1710^\circ)$

(46) write set $\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \frac{6}{7}\}$

(47) The English alphabet has 5 vowels and 21 consonant. How many words with 2 different vowels and 2 different consonants can be formed from alphabet

(48) A manufacturer has 600 litres of 12% solution of acid. How many litres of 30% acid solution must be added to it so that acid content in mixture will be more than 15% but less than 18%.

(49) The longest side of A is 3 times the shortest side and third side is 2 cm shorter than the longest side. If perimeter of A is atleast 61 cm find minumum length of shorter side

(50) How many litres of water will have to be added to 1125 litres of 45% solution of acid so that resulting mixture will contain more than 25% but less than 30% acid content